



2012 Half-Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Wednesday 16th May 2012

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.

Total — 99 Marks

- All questions may be attempted.

Section I – 11 Marks

- Questions 1–11 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 88 Marks

- Questions 12–19 are of equal value.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Collection

- Write your name, class and master clearly on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Twelve.
- Write your name and master on this question paper and submit it with your answers.

5A: DS

5D: DNW

5G: SO

5B: TCW

5E: LYL

5H: BR

5C: REP

5F: MLS

5I: SJE

Checklist

- SGS booklets — 8 per boy
- Multiple choice answer sheet
- Candidature — 134 boys

Examiner

DNW

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The value of $(3 - \sqrt{2})^2$ is:

- (A) 7 (B) 11 (C) $7 - 6\sqrt{2}$ (D) $11 - 6\sqrt{2}$

QUESTION TWO

The gradient of the line $4x - 3y + 2 = 0$ is:

- (A) $-\frac{4}{3}$ (B) $-\frac{3}{4}$ (C) $\frac{4}{3}$ (D) $\frac{3}{4}$

QUESTION THREE

Which of the following are the correct two factors of $x^3 + 64$?

- (A) $(x - 4)(x^2 + 4x + 16)$ (B) $(x + 4)(x^2 - 4x + 16)$
(C) $(x - 4)(x^2 + 4x - 16)$ (D) $(x + 4)(x^2 + 4x + 16)$

QUESTION FOUR

The domain of $y = \frac{1}{x^2 - 4}$ is:

- (A) $x = 2$ or $x = -2$ (B) $x \neq 2$ and $x \neq -2$
(C) $-2 < x < 2$ (D) $x < -2$ or $x > 2$

QUESTION FIVE

The exact value of $\operatorname{cosec} 240^\circ$ is:

- (A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$ (C) $\frac{2}{\sqrt{3}}$ (D) $-\frac{2}{\sqrt{3}}$

QUESTION SIX

The centre of the circle with equation $x^2 - 8x + y^2 + 2y = 3$ is:

- (A) $(4, -1)$ (B) $(8, -2)$ (C) $(-4, 1)$ (D) $(-8, 2)$

QUESTION SEVEN

The solution of $2x - x^2 < 0$ is:

- (A) $0 < x < 2$ (B) $x < 0$ or $x > 2$
(C) $x > 2$ (D) $x < 0$

Exam continues overleaf ...

QUESTION EIGHT

The solution of $|x - 2| = -1$ is:

- (A) $x = 1$ (B) $x = 3$
(C) $x = 3$ or 1 (D) there is no solution

QUESTION NINE

The limiting sum of the geometric series $6 - 3 + 1\frac{1}{2} - \dots$ is:

- (A) 4 (B) $4\frac{1}{2}$ (C) 5 (D) 12

QUESTION TEN

Let $A = (-1, -2)$ and $B = (5, 1)$. The point that divides AB in the ratio $2 : 1$ is:

- (A) $(0, -1\frac{1}{2})$ (B) $(1, -1)$ (C) $(2, -\frac{1}{2})$ (D) $(3, 0)$

QUESTION ELEVEN

The derivative of $3x^3 + 2x$ is:

- (A) $3x^2 + 2$ (B) $9x^2 + 2x$ (C) $9x^2 + 2$ (D) $3x^2$

————— End of Section I —————

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION TWELVE (11 marks) Use a separate writing booklet.

(a) Write down the exact value of:

(i) $\sec 60^\circ$

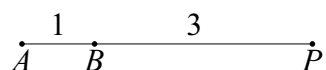
(ii) $\cot 30^\circ$

(iii) $\sin(-225^\circ)$

(b) Simplify $\sqrt{54} + \sqrt{24}$.

(c) Express $\frac{2}{3 - \sqrt{5}}$ in the form $a + b\sqrt{5}$, where a and b are rational.

(d)



Write down the ratio in which P divides the interval AB .

(e) Find the fifteenth term of the arithmetic sequence $5, 8, 11, \dots$

(f) Differentiate:

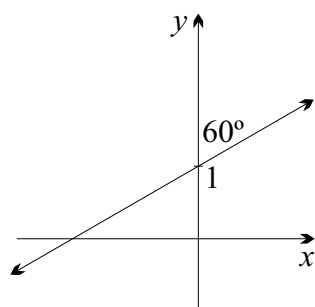
(i) $g(x) = x(x^2 + 1)$

(ii) $h(x) = \sqrt{x}$

(iii) $f(x) = \frac{3}{x^2}$

QUESTION THIRTEEN (11 marks) Use a separate writing booklet.

(a)



Determine the equation of the line shown in the diagram above.

Exam continues overleaf ...

QUESTION THIRTEEN (Continued)

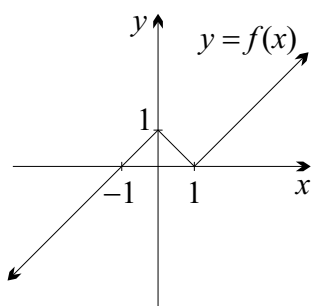
(b) Use a suitable series to express $0.\dot{2}\dot{1}$ as a fraction in simplest form.

(c) Solve $\frac{3}{x-1} \leq 2$.

(d) Given that $\sin \theta = -\frac{3}{7}$ and that $\tan \theta < 0$, find the exact value of $\cos \theta$.

QUESTION FOURTEEN (11 marks) Use a separate writing booklet.

(a)



The graph of $y = f(x)$ is shown above. Sketch the following on separate axes, showing all key features.

(i) $y = f(-x)$

(ii) $y = f(x+1)$

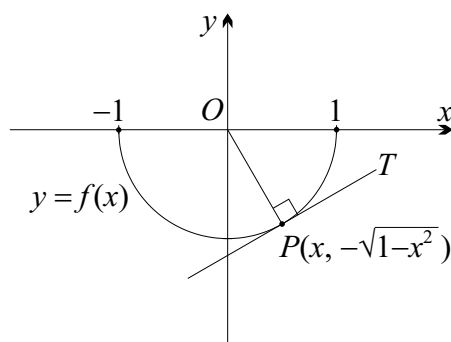
(b) Simplify $\frac{\tan \theta}{\sec \theta}$.

(c) Solve $\tan 2\theta = \frac{1}{\sqrt{3}}$, for $0^\circ \leq \theta < 360^\circ$.

(d) (i) Sketch $y = (x+2)(x-1)$, showing the intercepts with the axes.

(ii) Hence sketch $y = |(x+2)(x-1)|$ on a separate set of axes.

(e)



The graph of $y = f(x)$ is shown above, where $f(x) = -\sqrt{1-x^2}$. Use the fact that the tangent to a circle is perpendicular to the radius at the point of contact to find the derivative of $f(x) = -\sqrt{1-x^2}$.

QUESTION FIFTEEN (11 marks) Use a separate writing booklet.

- (a) Prove the identity:

$$\tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$$

- (b) In music, the frequencies $f_1, f_2, f_3 \dots$ of the notes in a chromatic scale form a geometric progression. It is known that the frequency of A below middle C is $f_1 = 220$ Hz. The frequency of A above middle C is $f_{13} = 440$ Hz. Find f_9 , the frequency of F above middle C . Round your answer to one decimal place.

- (c) (i) Find $\frac{dy}{dx}$ when $y = (x^2 - 1)^4$.

- (ii) Find $\frac{dy}{dx}$ when $y = x^2(x - 3)^5$. Give your answer in factored form.

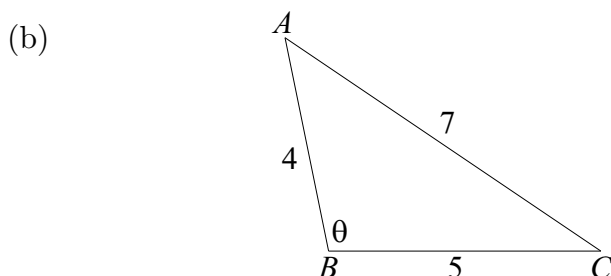
- (d) Let $f(x)$ be the function:

$$f(x) = \begin{cases} ax^2 & \text{when } x \leq 3 \\ \sqrt{25 - x^2} & \text{when } 3 < x \leq 5 \end{cases}$$

Given that $f(x)$ is continuous at $x = 3$, find the value of a .

QUESTION SIXTEEN (11 marks) Use a separate writing booklet.

- (a) What is the least number of terms of the series $50 + 43 + 36 + \dots$ that must be taken so that the sum is negative?



The diagram above shows $\triangle ABC$ where $AB = 4$, $BC = 5$, $AC = 7$ and $\angle ABC = \theta$.

- (i) Find the value of $\cos \theta$.
- (ii) Hence find the exact area of $\triangle ABC$.

- (c) Differentiate $f(x) = \left(x + \frac{1}{x}\right)^2$.

- (d) Determine the perpendicular distance between the lines

$$3x - y + 1 = 0 \quad \text{and} \quad 3x - y + 3 = 0.$$

Exam continues overleaf ...

QUESTION SEVENTEEN (11 marks) Use a separate writing booklet.

- (a) How many positive integer powers of 3 are less than 10^{12} ?
- (b) Use the quotient rule to show that the derivative of $y = \frac{x}{x^2 + 1}$ is $\frac{dy}{dx} = \frac{1 - x^2}{(x^2 + 1)^2}$.
- (c) (i) Determine the coordinates of the points where the line $y = x + 2$ intersects the parabola $y = x^2$.
- (ii) Sketch the intersection of the regions $y < x + 2$ and $y \geq x^2$. Pay careful attention to any corners.
- (d) The lines ℓ_1 and ℓ_2 intersect at P . The equations of the lines are:

$$\ell_1 : x + 2y + 3 = 0$$

$$\ell_2 : 3x + y - 2 = 0$$

Determine the equation of the line PQ where $Q = (-1, 2)$, without finding the coordinates of P .

QUESTION EIGHTEEN (11 marks) Use a separate writing booklet.

- (a) Let $y = f(x)$ where $f(x) = \frac{x}{x^2 - 1}$.
- (i) (α) Show that $f(x)$ is an odd function.
- (β) Solve $f(x) = 1$.
- (γ) Hence, or otherwise, solve $f(x) = -1$.
- (ii) Find any x -intercepts of $y = f(x)$.
- (iii) Find the equations of any asymptotes.
- (iv) Sketch the graph of $y = f(x)$ for $-4 \leq x \leq 4$, showing the features you have found.
- (b) Let $h(x) = \frac{1}{2} \left(1 + \frac{x}{|x|} \right)$, where $x \neq 0$.

Sketch $y = h(x)$ by considering separately $x < 0$ and $x > 0$.

QUESTION NINETEEN (11 marks) Use a separate writing booklet.

- (a) Let $f(x) = x^3 - 2x^2 + 1$. The function $g(x)$ is the result of shifting $f(x)$ to the right by k units. Given that the coefficient of x^2 in $g(x)$ is zero, find the value of k .
- (b) Consider the parabola with equation $y = -x^2 + 4x - 3$.
- (i) Find the equation of the tangent to this parabola at the point where $x = p$.
- (ii) Given that the tangent in part (i) passes through the origin, determine p .
- (c) Consider the sequence defined by:

$$x_1 = 3$$
$$x_n = \frac{(x_{n-1})^2 + 6}{2x_{n-1}} \text{ for } n \geq 2.$$

In the following you may assume that every term of this sequence is positive.

- (i) Write down the first three terms, simplifying any fractions.
- (ii) Show that $x_n - \sqrt{6} \geq 0$ for $n \geq 2$.
- (iii) Let $\epsilon_n = x_n - \sqrt{6}$.
- (α) Write down the exact value of ϵ_1 .
- (β) Show that $\frac{\epsilon_n}{\epsilon_{n-1}} < \frac{1}{2}$ for $n \geq 2$.
- (γ) Hence determine $\lim_{n \rightarrow \infty} x_n$.

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

SYDNEY GRAMMAR SCHOOL



2012
Half-Yearly Examination
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MATHEMATICS EXTENSION 1
Wednesday 16th May 2012

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS:MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Question Eleven

A ☐ B ☐ C ☐ D ☐

Q 1 (D)**Q 2** (C)**Q 3** (B)**Q 4** (B)**Q 5** (D)**Q 6** (A)**Q 7** (B)**Q 8** (D)**Q 9** (A)**Q 10** (D)**Q 11** (C)**QUESTION TWELVE** (11 marks)

(a) (i) $\sec 60^\circ = 2$



(ii) $\cot 30^\circ = \sqrt{3}$



(iii) $\sin(-225^\circ) = \frac{1}{\sqrt{2}}$



(b) $\sqrt{54} + \sqrt{24} = 3\sqrt{6} + 2\sqrt{6}$
 $= 5\sqrt{6}$



(c) $\frac{2}{3 - \sqrt{5}} = \frac{2}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}}$
 $= \frac{3 + \sqrt{5}}{2} \quad \text{or} \quad \frac{3}{2} + \frac{1}{2}\sqrt{5}$



(d) $-4 : 3 \quad \text{or} \quad 4 : -3 \quad \text{or} \quad \text{externally } 4 : 3$



(e) $T_{15} = 5 + 14 \times 3$
 $= 47$



(f) (i) $g'(x) = 3x^2 + 1$



(ii) $h'(x) = \frac{1}{2}x^{-\frac{1}{2}} \quad \text{or} \quad \frac{1}{2\sqrt{x}}$



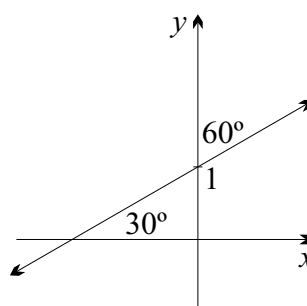
(iii) $f'(x) = -6x^{-3} \quad \text{or} \quad -\frac{6}{x^3}$

**Total for Question 12:** 11 Marks**QUESTION THIRTEEN** (11 marks)

(a) gradient = $\tan 30^\circ$

$= \frac{1}{\sqrt{3}}$

so $y = \frac{1}{\sqrt{3}}x + 1$



(b) $0.\dot{2}\dot{1} = 0.21 + 0.0021 + 0.000021 + \dots$



$$= \frac{0.21}{1 - \frac{1}{100}}$$



$$= \frac{21}{99}$$

$$= \frac{7}{33}$$



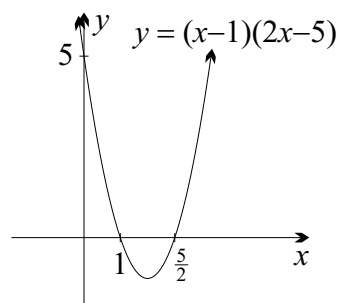
(c) $\frac{3}{x-1} \leq 2 \quad ; \quad x \neq 1$

so $3(x-1) \leq 2(x-1)^2$

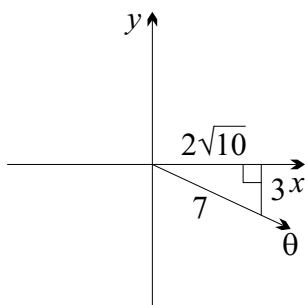
or $0 \leq 2(x-1)^2 - 3(x-1)$

thus $0 \leq (x-1)(2x-5) \quad ; \quad x \neq 1$

hence $x < 1 \quad \text{or} \quad x \geq \frac{5}{2}$



(d) Both $\sin \theta$ and $\tan \theta$ are negative so θ is in the 4th quadrant, and $\cos \theta > 0$.



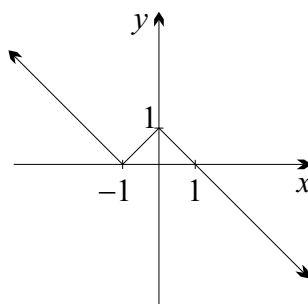
so $\cos \theta = \frac{2\sqrt{10}}{7}$



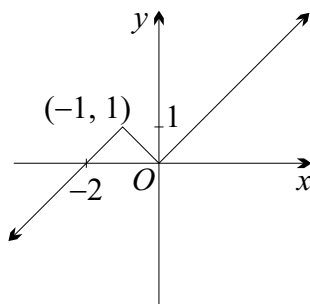
Total for Question 13: 11 Marks

QUESTION FOURTEEN (11 marks)

(a) (i)



(ii)



$$\begin{aligned} \text{(b)} \quad \frac{\tan \theta}{\sec \theta} &= \frac{\sin \theta}{\cos \theta \times \frac{1}{\cos \theta}} \\ &= \sin \theta \end{aligned}$$



$$\text{(c)} \quad \tan 2\theta = \frac{1}{\sqrt{3}} \quad 0^\circ \leq \theta < 360^\circ$$

Let $u = 2\theta$ then

$$\tan u = \frac{1}{\sqrt{3}} \quad 0^\circ \leq \theta < 720^\circ$$

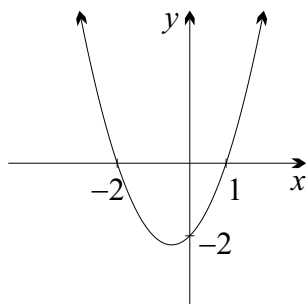
The related angle is 30° so

$$u = 30^\circ, 210^\circ, 390^\circ, 570^\circ$$

$$\text{thus} \quad \theta = 15^\circ, 105^\circ, 195^\circ, 285^\circ$$



(d) (i)

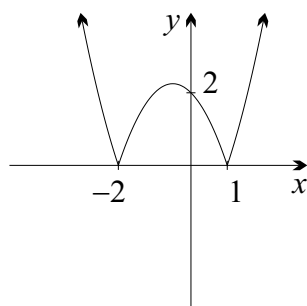


x -int

y -int



(ii)



$$\text{(e)} \quad \text{grad } OP = -\frac{\sqrt{1-x^2}}{x}$$

$$\text{so} \quad f'(x) = \frac{x}{\sqrt{1-x^2}}$$



Total for Question 14: 11 Marks

QUESTION FIFTEEN (11 marks)

$$\begin{aligned}
 \text{(a)} \quad \text{LHS} &= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} & \checkmark \\
 &= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} \\
 &= \frac{1}{\cos \theta \sin \theta} \quad (\text{by Pythagoras}) & \checkmark \\
 &= \text{RHS}
 \end{aligned}$$

[Do not penalise the lack of a reason above.]

$$\begin{aligned}
 \text{(b)} \quad a &= 220 \\
 \text{and } ar^{12} &= 440 & \checkmark \\
 \text{so } r &= \sqrt[12]{2} \quad (\text{positive only since all frequencies are positive}) & \checkmark \\
 [\text{Do not penalise a negative answer so long as the positive is also given.}] \\
 f_9 &= 220 \times 2^{\frac{2}{3}} \\
 &\doteq 349.2 \quad (\text{correct to 1 decimal place}) & \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) (i)} \quad y &= (x^2 - 1)^4 \\
 \text{Let } u &= x^2 - 1 \\
 \text{so } u' &= 2x \\
 \text{and } y &= u^4. & \checkmark \\
 \text{Thus } \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\
 &= 4u^3 \times u' \\
 &= 8x(x^2 - 1)^3 & \checkmark
 \end{aligned}$$

[Accept an answer with no working only if it is completely correct.]

$$\begin{aligned}
 \text{(ii)} \quad y &= x^2(x - 3)^5 \\
 &= uv \\
 \text{where } u &= x^2 \quad \text{so } u' = 2x \\
 \text{and } v &= (x - 3)^5 \quad \text{so } v' = 5(x - 3)^4. & \checkmark \\
 y' &= uv' + u'v \\
 &= x^2 5(x - 3)^4 + 2x(x - 3)^5 & \checkmark \\
 &= x(x - 3)^4(7x - 6) & \checkmark
 \end{aligned}$$

- (d) For continuity at $x = 3$ we require $\lim_{x \rightarrow 3^-} f(x) = f(3) = \lim_{x \rightarrow 3^+} f(x)$.

Now $f(3) = 9a$

and $\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \sqrt{25 - x^2}$
 $= 4$

Thus $9a = 4$

or $a = \frac{4}{9}$



Total for Question 15: 11 Marks

QUESTION SIXTEEN (11 marks)

- (a) For the arithmetic sequence $a = 50$ and $d = -7$, so:

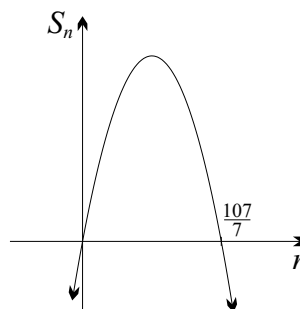
$$S_n = \frac{1}{2}n(107 - 7n)$$

Thus we need to solve

$$\frac{1}{2}n(107 - 7n) < 0$$

hence $n > \frac{107}{7}$ or $15\frac{2}{7}$

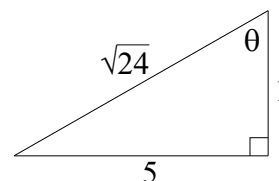
Thus $n = 16$



- (b) (i) Now $7^2 = 4^2 + 5^2 - 2 \times 4 \times 5 \cos \theta$ (by the cosine rule)

so $\cos \theta = -\frac{1}{5}$

(ii) Area $= \frac{1}{2} \times 4 \times 5 \times \sin \theta$
 $= \frac{1}{2} \times 4 \times 5 \times \frac{\sqrt{24}}{5}$
 $= 4\sqrt{6}$ square units



- (c) Expanding $f(x) = x^2 + 2 + x^{-2}$

so $f'(x) = 2x - 2x^{-3}$ or $2x - \frac{2}{x^3}$



- (d) The point $(0, 3)$ is on the second line (y -intercept),
 so the distance d to the first line is:

$$d = \frac{|3 \times 0 - 1 \times 3 + 1|}{\sqrt{3^2 + (-1)^2}}$$

$$= \frac{\sqrt{10}}{5}$$



Total for Question 16: 11 Marks

QUESTION SEVENTEEN (11 marks)

- (a) That is, find the largest n such that

$$3^n < 10^{12}$$

Taking logs of both sides

$$n \log_{10} 3 < 12$$

$$\text{or} \quad n < \frac{12}{\log_{10} 3} \\ < 25.15 \dots$$

$$\text{thus} \quad n = 25$$



(b)
$$y = \frac{x}{x^2 + 1} \\ = \frac{u}{v}$$

where $u = x$ so $u' = 1$

and $v = x^2 + 1$ so $v' = 2x$

Then
$$\frac{dy}{dx} = \frac{u'v - uv'}{v^2} \\ = \frac{1 \times (x^2 + 1) - x \times 2x}{(x^2 + 1)^2} \\ = \frac{1 - x^2}{(x^2 + 1)^2}$$



- (c) (i) That is, solve

$$x^2 = x + 2$$

so $x^2 - x - 2 = 0$

or $(x - 2)(x + 1) = 0$

thus $x = -1$ or 2

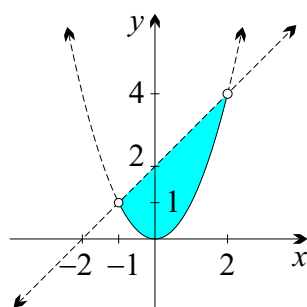
and $y = 1$ or 4

That is: $(-1, 1)$ and $(2, 4)$.

[Do not penalise failure to find y -values.]



- (ii)



The regions is below the line,
above the parabola,
and excludes the corners and line.



(d) The general equation of the line through P is

$$(x + 2y + 3) + k(3x + y - 2) = 0$$

So at $Q(-1, 2)$ this gives

$$6 - 3k = 0$$

so

$$k = 2$$

Thus the equation of PQ is

$$(x + 2y + 3) + 2(3x + y - 2) = 0$$

or

$$7x + 4y - 1 = 0$$

Total for Question 17: 11 Marks

QUESTION EIGHTEEN (11 marks)

(a) For $y = f(x)$ where $f(x) = \frac{x}{x^2 - 1}$:

$$\begin{aligned} \text{(i) } (\alpha) \quad f(-x) &= \frac{(-x)}{(-x)^2 - 1} \\ &= -\frac{x}{x^2 - 1} \\ &= -f(x) \end{aligned}$$

$$\text{(}\beta\text{) That is } \frac{x}{x^2 - 1} = 1$$

$$\text{so } x^2 - x - 1 = 0$$

$$\text{or } x = \frac{1 \pm \sqrt{5}}{2}$$

$$\text{(}\gamma\text{) Since } f(x) \text{ is odd, the values are } x = -\frac{1 \pm \sqrt{5}}{2}.$$

(ii) $f(0) = 0$. There are no other intercepts.

$$\begin{aligned} \text{(iii) } \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1 - \frac{1}{x^2}} \\ &= 0 \end{aligned}$$

hence also, by the oddness of $f(x)$,

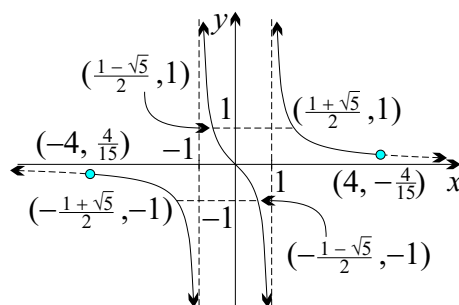
$$\lim_{x \rightarrow -\infty} f(x) = 0.$$

So the x -axis is an asymptote (or $y = 0$)

There are vertical asymptotes at $x = -1$ or 1

where the denominator is zero and the numerator is not zero.

(iv)

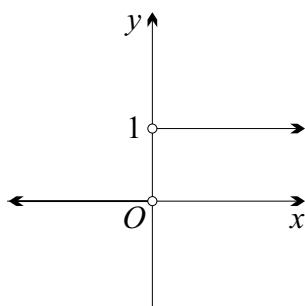


odd function

shape

[Do not penalise failure to show coordinates where $y = \pm 1$.]

$$\begin{aligned}
 \text{(b)} \quad h(x) &= \frac{1}{2} \left(1 + \frac{x}{|x|} \right) \\
 &= \begin{cases} \frac{1}{2} \left(1 + \frac{x}{-x} \right) & \text{when } x < 0 \\ \frac{1}{2} \left(1 + \frac{x}{x} \right) & \text{when } x > 0 \end{cases} \\
 &= \begin{cases} 0 & \text{when } x < 0 \\ 1 & \text{when } x > 0 \end{cases}
 \end{aligned}$$



Total for Question 18: 11 Marks

QUESTION NINETEEN (11 marks)

(a)
$$\begin{aligned} g(x) &= f(x - k) \\ &= (x - k)^3 - 2(x - k)^2 + 1 \\ &= x^3 - (3k + 2)x^2 + (3k^2 + 4k)x + (1 - 2k^2 - k^3) \end{aligned}$$



so $3k + 2 = 0$

thus $k = -\frac{2}{3}$



(b) For $y = -x^2 + 4x - 3$:

(i) $y' = -2x + 4$

so at $x = p$ $y' = 4 - 2p$

and $y = -p^2 + 4p - 3$

Thus the equation of the tangent is:

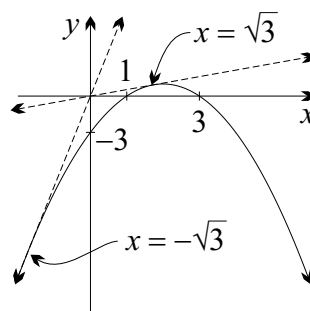
$$y + p^2 - 4p + 3 = (4 - 2p)(x - p)$$

(ii) The point $(0, 0)$ is on the tangent so:

$$p^2 - 4p + 3 = 2p^2 - 4p$$

so $p^2 = 3$

or $p = -\sqrt{3}$ or $\sqrt{3}$



(c) (i) $x_1 = 3$

$$x_2 = \frac{9+6}{6} = \frac{5}{2}$$

$$x_3 = \frac{\frac{25}{4}+6}{5} = \frac{49}{20}$$



(ii)
$$\begin{aligned} \text{LHS} &= x_n - \sqrt{6} \\ &= \frac{(x_{n-1})^2 + 6}{2x_{n-1}} - \sqrt{6} \quad \text{for } n \geq 2 \\ &= \frac{(x_{n-1})^2 - 2x_{n-1}\sqrt{6} + 6}{2x_{n-1}} \\ &= \frac{(x_{n-1} - \sqrt{6})^2}{2x_{n-1}} \end{aligned}$$



Now the numerator is positive or zero and the denominator is positive hence $\text{LHS} \geq 0$.

(iii) (α) $\epsilon_1 = 3 - \sqrt{6}$



(β)
$$\begin{aligned} \frac{\epsilon_n}{\epsilon_{n-1}} &= \frac{x_n - \sqrt{6}}{x_{n-1} - \sqrt{6}} \\ &= \frac{(x_{n-1} - \sqrt{6})^2}{2x_{n-1}} \times \frac{1}{(x_{n-1} - \sqrt{6})} \\ &= \frac{x_{n-1} - \sqrt{6}}{2x_{n-1}} \\ &< \frac{x_{n-1}}{2x_{n-1}} \quad (\text{increased numerator}) \\ &< \frac{1}{2}. \end{aligned}$$



(γ) Since $\frac{\epsilon_n}{\epsilon_{n-1}} < \frac{1}{2}$ and $\epsilon_n \geq 0$,

every term of the sequence ϵ_n is no larger than the corresponding term of the geometric progression

$$(3 - \sqrt{6}), \frac{1}{2}(3 - \sqrt{6}), \frac{1}{4}(3 - \sqrt{6}), \dots$$

That is $0 \leq \epsilon_n \leq (3 - \sqrt{6})\left(\frac{1}{2}\right)^n$

so $0 \leq \lim_{n \rightarrow \infty} \epsilon_n \leq \lim_{n \rightarrow \infty} (3 - \sqrt{6})\left(\frac{1}{2}\right)^n$

and $0 \leq \lim_{n \rightarrow \infty} \epsilon_n \leq 0$.

Thus $\lim_{n \rightarrow \infty} \epsilon_n = 0$

so $\lim_{n \rightarrow \infty} x_n - \sqrt{6} = 0$

hence $\lim_{n \rightarrow \infty} x_n = \sqrt{6}$.



Total for Question 19: 11 Marks